

A preliminary assessment of CO₂ capture, transport, and storage network for China's steel sector

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ABSTRACT

Carbon capture and storage (CCS) has emerged as a promising way for achieving deep decarbonization of China's steel sector. Large-scale CCS deployment necessitates a comprehensive source-sink matching analysis to offer early engineering guidance. To address this issue, this study conducted a source-sink matching analysis of China's steel sector, assessing CO₂ sources, CO₂ storage capacities and costs, and CO₂ pipeline layouts under three capture targets. The findings reveal an obvious spatial mismatch between CO₂ sources and potential CO₂ sinks for China's steel sector. CO₂ sources are mainly situated in eastern China, while main CO₂ storage sites are located in western China. The average unit costs of CCS under low (30%), medium (60%), and full capture (90%) targets are 83, 91, and 118 \$/t CO₂, respectively. In particular, as capture target increases, the CO₂ transportation cost will supersede the capture cost and dominate the average CCS unit cost, owing to the construction of longer pipelines. This study recommends prioritizing CCS demonstration with steel plants in regions having suitable CO₂ storage basins nearby. Moreover, future investigations should concentrate on on-site experiments of CO₂ storage basins and the technology development of offshore carbon storage in China.

1. Introduction

Carbon capture and storage (CCS) is anticipated to play a crucial role in the decarbonization of China's steel sector. As the world's largest steel producer, China's steel sector contributes 57% of global steel production (World Steel Association, 2021) and is responsible for 20% of China's total CO₂ emissions (Yang et al., 2020). Several strategies can be used to mitigate CO₂ emissions from China's steel sector, including energy efficiency improvements, demand-side steel management, and innovative technologies (Lin et al., 2021). As estimated by the International Energy Agency (International Energy Agency, 2020a), innovative technologies, particularly hydrogen-based steelmaking and CCS technologies, are expected to contribute to over 20% of CO₂ reduction of global steel sector. In comparison to hydrogen-based steelmaking technologies, CCS offers a direct and effective way of reducing CO₂ emissions with fewer alterations to existing processes. Given the coal-dominated energy

structure in China's steel sector, CCS is recognized as a pivotal and promising technology that can strike a balance between industry development and CO₂ reduction in such a hard-to-abate sector (International Energy Agency, 2020b).

CCS has gained increasing attention in China's carbon neutrality policy framework following China's commitment to achieving carbon neutrality by 2060. Its formal recognition in China's National Science and Technology Development Plan (China TSCO, 2006–2020) in 2006 marked a significant milestone, marking CCS as a leading-edge technology pivotal to facilitating the low-carbon transition of fossil energy. Since then, China has progressively pursued theoretical research, technological development, and small-scale engineering demonstrations of CCS. In 2019, the Ministry of Science and Technology released China's CCS technology development roadmap (Ministry of Science and Technology of the People's Republic of China, 2019), outlining the goals and priorities across various development stages. In 2020, China officially

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committed its national climate goal of achieving carbon neutrality by 2060 (Ministry of Foreign Affairs of the People's Republic of China), further heightened the spotlight on CCS. In 2021, CCS technology was incorporated in China's 14th Five-Year Plan (People's Government of China) for the first time, a pivotal document guiding the nation's economic and social development. Consequently, both national and local governments issued a series of policy documents aimed at actively managing CCS technology research and development, setting technical standards, and offering financing support (Xinhua Net, 2030; Xinhua Net). Furthermore, there has been an increasing emphasis on the application of CCS within industry sectors, particularly the steel and cement sectors, which are often regarded as typical hard-to-abate industry sectors (Ministry of Industry and Information Technology of the People's Republic of Chinaa; Ministry of Industry and Information Technology of the People's Republic of Chinab). However, on a global scale, CCS implementation in the steel sector remains largely in the early stage of technological readiness. Presently, the most notable near-commercial-scale CCS project in a steel plant is located in the United Arab Emirates, where CCS technology is deployed in a gas-based direct reduction steel plant (International Energy Agency, 2020a). In China, CCS initiatives in the steel sector primarily exist in the early stages of technology verification and small-scale technology demonstration (Xian et al., 2023).

Large-scale deployment of CCS requires detailed source-sink matching studies of CCS to provide early options and inform decision-making. CCS involves the process of capturing CO₂ from large point sources, then compressing and transporting it to a suitable storage location (International Energy Agency, 2020b). The focus of source-sink matching of CCS includes: 1) determining the amounts of captured CO₂ at various sources; 2) identifying suitable storage locations and quantifying storage injectivities and capacity; and 3) designing the siting of CO₂ transportation infrastructure and associated costs (Zhang et al., 2022a). Some source-sink matching models have been developed to facilitate CCS development around the world. For example, the InfraCCS model, developed by Morbee et al. (2012), was used to determine the potential layout and optimized cost of a Europe-wide CO₂ pipeline network. The SimCCS model, developed by Middleton R.S. et al. (Middleton and Bielicki, 2009; Middleton et al., 2012), was used to optimize the layout and costs of CO₂ pipelines in the US, allowing CO₂ pipelines merging and branching to efficiently match multiple CO₂ sources and sinks. It is noteworthy that the SimCCS model is open source to the public and undergoes continuous updates. Its latest version has integrated several new features, such as refined optimization models coupled with geographic information system (GIS), novel techniques for candidate network generation, and integration with high-performance computing platforms (Middleton et al., 2020a). For China, the ChinaCCUS-DSS model developed by Chen et al. (Sun and Chen, 2013, 2017, 2022; Tang et al., 2023), as well as the CO₂ pipeline network planning model by Wei et al. (2022a), can be used to explore cost-effective CO₂ pipeline layouts for CCS under China's carbon neutrality vision. However, these models designed for China mostly use straight lines to connect CO₂ sources and sinks, which is quite different from the actual pipeline construction situation.

Most previous studies of China's CCS source-sink matching analysis have concentrated on the power sector, leaving a vital knowledge gap in the detailed source-sink matching analysis for China's steel sector. Historically, CO₂ reduction studies in China have primarily centered on its coal-dominated energy production system, with CCS technology highlighted as a pivotal solution for achieving deep decarbonization in the coal-fired power generation sector (Jiang and Ashworth, 2021). For example, Fan et al., 2020, 2021a, 2021b established a comprehensive dataset of CO₂ sources and CO₂ storage sites to evaluate the CCS potential in China's coal-fired power sector. Wang et al. (2020) provided a cost-optimized CCS layout that identified CCS cluster opportunities for China's power sector. Wei et al. (2021) presented a cost curve of CCS projects in China's coal power sector. Tang et al. (2021) assessed a CCS

pipeline network for China's power sector toward the carbon neutrality vision. Conversely, the steel sector, as a significant energy consumer, has received less attention in CO₂ reduction, with early focus directed towards improving energy efficiency of production equipment to reduce costs associated with energy consumption. Furthermore, most steel enterprises actively engage in market competition and exhibit a high sensitivity to costs, diminishing their interest in adopting CCS technology due to its immaturity and high expense. However, interest in CCS in China's steel sector has grown since China's commitment to carbon neutrality in 2060 (Jiutian et al., 2022; Yang et al., 2022). For example, Ren et al. (2021) undertook a life-cycle cost analysis of CCS technology in a typical Chinese steel plant. Wei Ning et al. (Wei et al., 2022b) developed a cost curve for potential CCS projects in China's steel sector. Nevertheless, the existing body of literature just integrate CCS technology into energy system models as a decarbonization option (Wang et al., 2022; Zhang et al., 2022b), leaving a research gap in detailed CCS source-sink matching analysis for China's steel sector, particularly concerning the optimal layout of CO₂ pipelines (Desport and Selsosse, 2022).

To address this issue, this study undertook a detailed source-sink matching analysis of CCS in China's steel sector, aiming to offer early engineering guidance for CCS deployment in China's steel sector. This study built a comprehensive dataset containing CO₂ sources of 227 blast furnace/basic oxygen furnace (BF/BOF) steel plants and 12 prospective deep saline aquifers for CO₂ geological storage. Subsequently, preliminary siting of CO₂ pipeline network connecting CO₂ sources to sinks, while minimizing the average CCS cost, is executed using the open-source tool SimCCS (Middleton and Bielicki, 2009; Middleton et al., 2020a). Through this investigation, three pivotal inquiries regarding CCS in China's steel sector can be addressed:

- 1) How much CO₂ can be captured from China's steel plants, and what is the geological distribution across the country?
- 2) Where are suitable locations for storing the captured CO₂, and what is the maximum capacity for CO₂ storage?
- 3) What is the possible siting of CO₂ pipeline network connecting CO₂ sources to sinks, and what are the associated costs?

This is the first study to provide a high-resolution quantitative assessment of CO₂ capture, transport, and storage networks for China's steel sector. This comprehensive assessment includes quantitative analyses of the geographical distribution of CO₂ sources and sinks, preliminary siting of CO₂ transportation pipelines connecting sources and sinks, and overall costs and cost structure associated with CCS deployment in China's steel sector. In particular, this study integrated GIS techniques to conduct a high-resolution (nominally less than 1 km) cell-by-cell assessment of CO₂ pipeline routing and construction costs. This leads to the identification of more realistic least-cost path for CO₂ transportation networks than simplistic linear connections used in other studies. The findings of this study highlight the significant impact of CO₂ transportation expenses on the overall cost of CCS, given the locations of China's steel CO₂ sources relative to the most prospective of CO₂ storage basins. This work serves as a quantitative basis for providing early engineering guidance for prospective deployment of CCS in China's steel sector.

2. Methods and data

2.1. CO₂ sources

China's steel plants can be categorized into two main types according to different steel production routes: blast furnace/basic oxygen furnace (BF/BOF) and electric arc furnace (EAF) steel plants. This study primarily focused on the assessment of CO₂ emissions from BF/BOF steel plants, due to the understanding that the majority of CO₂ emissions in EAF steel plants originate from external electricity generation, limiting

the potential of on-site CO₂ capture.

Plant-level CO₂ emissions. This study evaluates CO₂ emissions from 227 BF/BOF steel plants in China for the year 2020. Initially, a detailed steel dataset containing 481 operating steel plants in China was acquired from publicly available sources, including information on geographic locations, steel production capacity, and steel production process for each plant (Global Energy Monitor). Then, data on electric arc furnace (EAF) steel plants were excluded from consideration. Subsequently, in the absence of detailed plant-level steel production data, provincial-level steel production data were utilized together with the aforementioned plant-level production capacity data to estimate CO₂ emissions for each steel plant. Subsequent to these adjustments, the CO₂ emissions of each plant were computed using the average direct CO₂ emissions of 1.4 tons per ton of crude steel (Lin et al., 2023). Ultimately, this study estimated CO₂ emissions for each of 227 blast BF/BOF steel plants for use in source-sink matching analysis, the dataset of which has been provided in the supplementary file.

Capture cost. In this study, post-combustion capture via chemical absorption is chosen as the method for CO₂ capture, given its well-established development and successful commercial demonstrations in real cases (McLaughlin et al., 2023; Arasto et al., 2013). This study adopted the cost function developed by Gunawan et al. (2024) to evaluate CO₂ capture costs, which are closely related to both the amount and concentration of CO₂, as listed in Eq. (1).

$$\text{Capture cost (\$/tCO}_2\text{)} = C_0 \cdot (\text{Mt CO}_2\text{ Captured})^{-b} \quad (1)$$

where the cost factor C_0 and b are closely related to the CO₂ concentration in captured flue gas, as listed in Table 1.

Regarding the CO₂ concentration in the captured flue gas, this study calculated an average CO₂ concentration of 20%, obtained through tracing the CO₂ flow within a steel plant (Fig. 1) and the CO₂ concentration in various steel production processes (Table 2). Subsequently, the cost factor C_0 and b in Eq. (1) can be determined through linear interpolation from Table 1.

Thus, the CO₂ capture cost of each steel plant can be calculated based on the estimated CO₂ emissions and the cost factor in Eq. (1). Detailed dataset on CO₂ capture costs of 227 BF/BOF steel plants are provided in the supplementary file.

2.2. CO₂ sinks

The geological formations suitable for CO₂ storage generally include deep saline aquifers, depleted oil or gas fields, and deep unmineable coal beds (Ajayi et al., 2019; Dahowski et al., 2009). In this study, deep saline aquifers are selected as geological sites for long-term CO₂ storage due to their large estimated storage potential (Jafari et al., 2017; Zhong et al., 2023).

CO₂ storage capacity and costs. A dynamic method was adopted to estimate CO₂ storage capacities and associated costs. Unlike conventional approaches in prior studies that assessed the storage capacity using static parameters such as area (m²), thickness (m), and storage efficiency factor (%) (Bachu et al., 2007), recent geological studies emphasize the significance of dynamic factors in determining storage capacity. Dynamic elements, including the CO₂ injection rate, CO₂ plume characteristics, and the configuration of CO₂ injection wells, play crucial roles in influencing storage capacity estimates and have been shown to yield different results compared to static estimates (Ranaee et al., 2022). To mitigate uncertainties in capacity estimates, this study

adopted a dynamic model developed by Middleton et al., 2020b, 2020c. This model incorporates input parameters that includes storage formation lateral area (m²), depth (m), thickness (m), porosity (%), permeability (mD), and geothermal gradient (°C/km). These parameters are utilized to calculate CO₂ plume migration assuming an idealized cylindrical reservoir with uniform geophysical properties, on the basis of which the storage capacity, aggregate injectivity, and number of individual injection wells that avoids plumes interaction (Fig. 2) is estimated, as well as associated CO₂ storage costs.

CO₂ storage site selection. Data of 12 onshore deep saline aquifers were chosen for the estimation of CO₂ capacity (Table 3). Between 2010 and 2012, China undertook a comprehensive nationwide assessment to ascertain the suitability of major sedimentary basins for geological CO₂ storage (The China Geological Survey Center for Hydrogeology and Environmental Geology Survey, 2017). Several studies further evaluated the potential of deep saline aquifers within these basins (Fan et al., 2021a; Wei et al., 2013). Drawing on insights gained from these assessments, this study selected 12 onshore deep saline aquifers to consider for CO₂ storage, taking into consideration the distribution of CO₂ sources originating from China's steel sector. Notably, this study didn't take into account offshore CO₂ storage, primarily due to the relatively high cost of offshore CO₂ storage technology compared to its onshore counterpart in China (Li et al., 2011). Detailed dataset of the CO₂ storage capacity and cost of 12 onshore deep saline aquifers are provided in the supplementary file.

2.3. CO₂ transportation via pipelines

Captured CO₂ can be transported from steel plants to storage sites via three main ways: truck, ship, and pipeline. While trucks and ships are typically used for small-scale CO₂ transportation, this study opts for CO₂ pipeline transportation due to its advantages in facilitating large-scale and long-distance CO₂ transportation.

CO₂ transportation costs. In this study, GIS maps were integrated to provide a comprehensive representation of actual geographic, topographic, and demographic features that would be considered when constructing onshore CO₂ pipelines (Fig. 3). Costs of CO₂ pipelines were then assessed on a cell-by-cell basis of selected GIS maps. Each cell was assigned a cost factor, signifying the relative expense associated with constructing CO₂ pipelines in that specific location. A value of 1 represents the standard pipeline construction cost, which itself is a function of diameter. The standard pipeline construction cost is calculated utilizing the cost model of Morgan et al. (2022), as modified by Wei et al. for China's (Wei et al., 2016). Cost factors less than 1 indicate areas where pipelines could be more easily constructed, such as alongside existing oil and gas pipelines, while values greater than 1 indicate higher construction costs, which especially characterize densely populated regions. Subsequently, these GIS maps were collectively weighted to create a cost surface. The resolution of the cost surface is 240×240 m. The GIS maps and the cost factors for weighting CO₂ pipeline construction costs are provided in the supplementary file.

Based on the constructed cost surface, the next step is to put the geographical coordinates of CO₂ sources and sinks into it, thereby optimizing the least-cost paths to connect CO₂ capture sources at steel plants with prospective CO₂ storage basins. The least-cost-path optimization problem has been extensively addressed in prior studies, and this analysis utilized the open-source software developed by Middleton et al. (Middleton et al., 2020a; Hoover et al., 2020) known as SimCCS. SimCCS is designed to aggregate CO₂ flows between sources and sinks into trunk pipelines, optimizing the utilization of economies of scale (Middleton and Bielicki, 2009; Middleton et al., 2020a) and determine the overall least-cost spatial layout for a known set of CO₂ sources and sinks in a geographic region of interest.

Table 1
The cost factors for the cost estimation of CO₂ capture.

CO ₂ (Vol %)	5%	10%	15%	60%
C_0	121.83	98.75	88.54	86.37
b	0.155	0.186	0.204	0.210

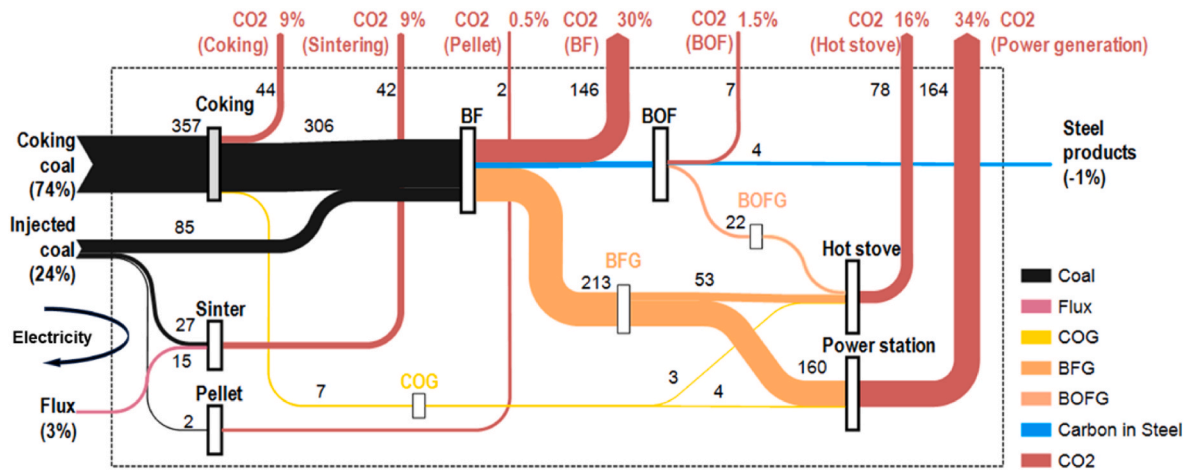
Fig. 1. CO₂ flows in a typical steel plant of China.

Table 2

The CO₂ flow share and CO₂ concentration in various steel production processes.

	Coking	Sintering	Pellet	BF	BOF	Hot stoven	Power generation	Average
CO ₂ flow share (% of the total) (Fig. 1)	9%	9%	0.5%	30%	1.5%	16%	34%	–
CO ₂ concentration (vol %) (Li et al., 2022)	18%	6%	3%	25%	15%	10%	24%	20%

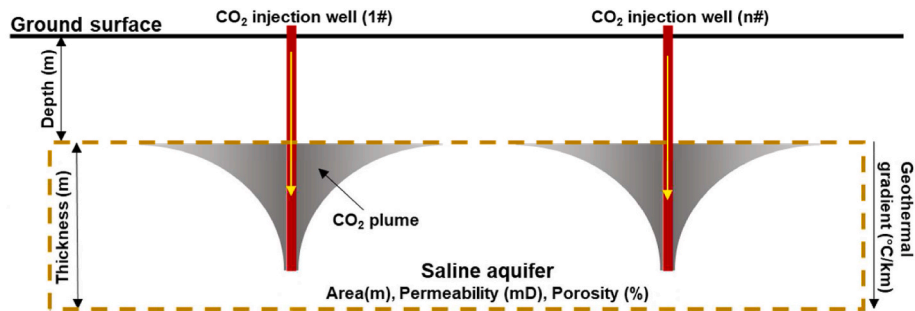
Fig. 2. Schematic diagram of CO₂ injection into the deep saline aquifer.

Table 3

Parameters for the estimation of CO₂ storage capacity of 12 deep saline aquifers in China (Jafari et al., 2017; Zhong et al., 2023; Ranaee et al., 2022; Dahowski et al., 2012).

	Depth (m)	Thickness (m)	Permeability (mD)	Porosity (%)	Geothermal gradient (°C/km)	Area (km ²)
Tarim Basin	4000	300	100	0.15	20	100000
Junggar Basin	1500	300	75	0.15	21	19000
Qaidam Basin	500	50	5	0.15	30	15000
Ordos Basin	1000	200	125	0.18	30	35000
Erlian Basin	1000	300	1000	0.15	40	16000
Hailaer Basin	2000	100	10	0.15	30	11000
Songliao Basin	1200	150	200	0.2	42	38000
Bohai Bay Basin	1500	200	800	0.23	55	30000
Southern North China	1000	300	500	0.2	35	52000
Subei Basin	1000	300	5	0.2	30	10000
Jiangnan Basin	1000	150	10	0.1	35	2000
Sichuan Basin	3500	400	5	0.1	23	50000

3. Results

This section presents key results of the source-sink matching analysis of China's steel sector. Fig. 4 maps the BF/BOF CO₂ sources and the 12 identified saline aquifer storage sites. Fig. 5 maps the cost-optimized CO₂ pipeline networks connecting CO₂ sources to sinks for the three steel-sector capture targets, and Fig. 6 shows the corresponding average

unit cost for CO₂ capture, transport, and storage. Fig. 7 presents the regional disparities on the average unit cost of CCS in three selected CO₂ cluster.

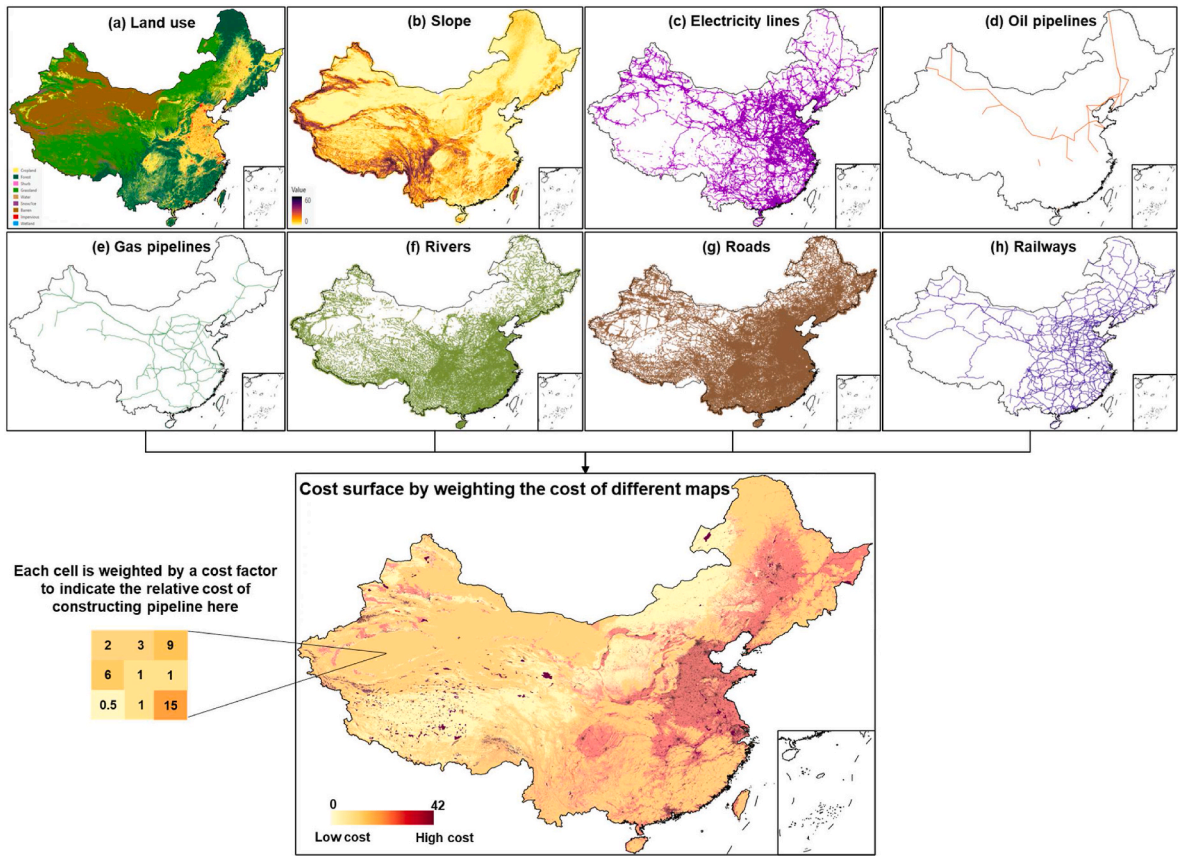


Fig. 3. Cost surface reflecting weightings of CO₂ pipeline construction costs according to map features depicted in panels. (The resolution of the land-use map is 720×720 m. The resolution of other input maps is 240×240 m. The resolution of the cost surface is 240×240 m. The cost factors in the cost surface are the cost of constructing a CO₂ pipeline relative to the cost for a reference standard pipeline. A value less than 1 means that pipelines in that cell cost less than the standard. A value larger than 1 means higher construction costs.)

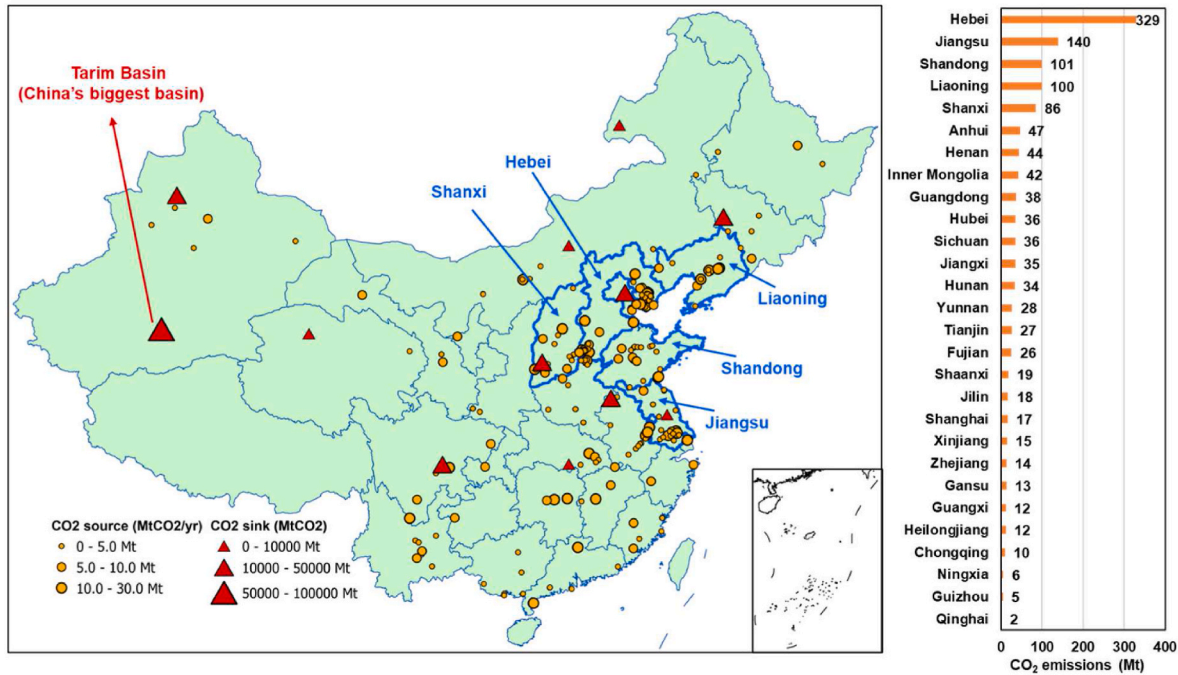


Fig. 4. Spatial distribution of CO₂ sources and potential CO₂ sinks for China's steel sector and estimated province-level CO₂ emissions in 2020 associated with BF/BOF steel production.

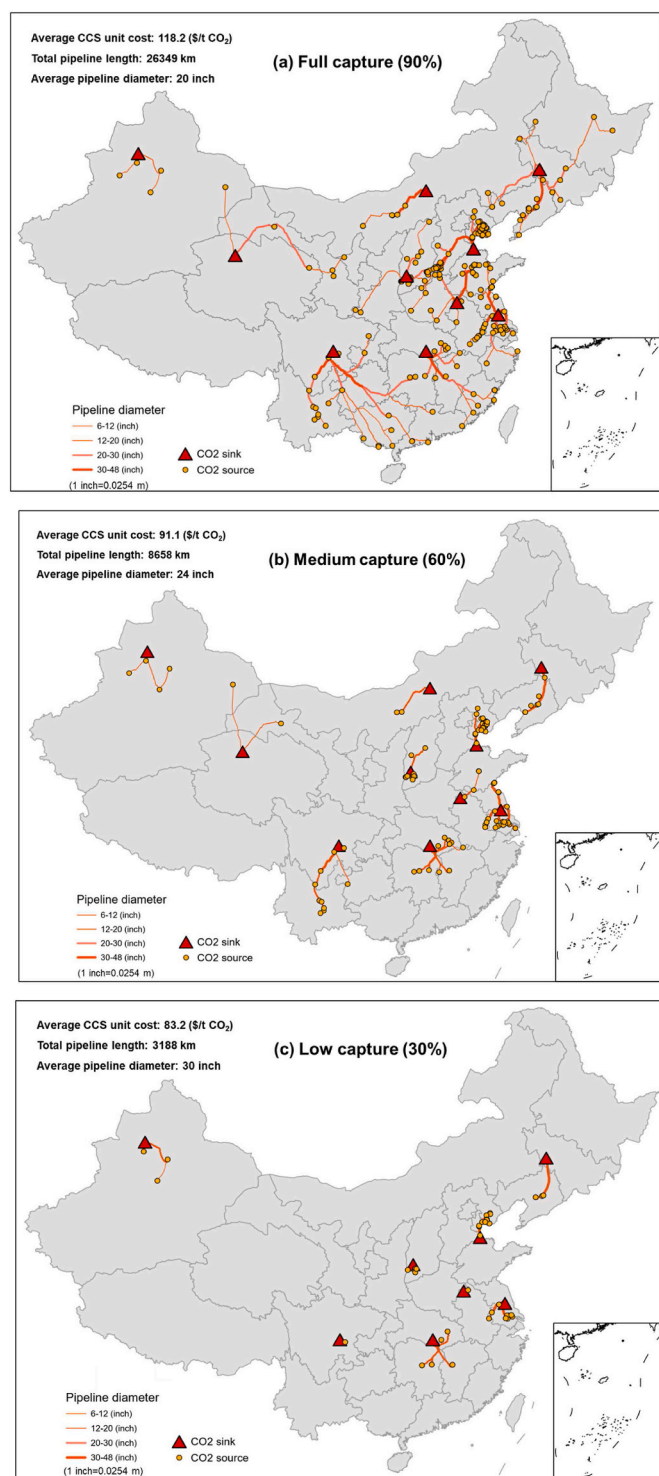


Fig. 5. The cost-optimized CO₂ pipeline network under three CO₂ capture targets (Given that the CO₂ capture rate set in this study is 90%, the full capture target refers to a 90% of total CO₂ reduction, not 100%).

3.1. The spatial distribution of CO₂ sources and potential CO₂ sinks for China's steel sector

An evident spatial mismatch and relatively long transportation distances are observed between CO₂ sources and potential CO₂ sinks for China's steel sector. The CO₂ sources originating from China's steel plants are predominantly situated in eastern China, while the primary CO₂ storage sites are located in western China (Fig. 4).

CO₂ sources are concentrated along the coastal areas extending from northeast to east China. i.e., in China's main economic hub convenient to maritime transportation that facilitates the movement of inputs and products of steel producers. Hebei Province stands out as China's largest steel-producing province, producing 250 million tons of crude steel in 2020, accounting for 25% of China's total production. CO₂ emissions from steel production in Hebei Province reached 329 million tons in 2020, more than double those of the second-ranking province, Jiangsu.

Most of China's major basins suitable for CO₂ geological storage are situated in the western regions of the country. The Tarim Basin, the largest among them, is located in Xinjiang Province in western China, considerably distant from coastal regions where CO₂ sources are concentrated. Although there are some available basins in the east in closer proximity to CO₂ sources, many of them are situated near densely populated areas, introducing significant uncertainties regarding the availability of CO₂ storage sites and the viability of constructing CO₂ transportation infrastructure. A detailed discussion of these challenges is presented in subsequent sections.

3.2. The siting and associate costs of CO₂ pipeline network connecting CO₂ sources to sinks under three steel-sector capture targets

This section presents the siting of cost-optimized CO₂ pipeline network connecting CO₂ sources to sinks, including the length and diameter (Fig. 5), and then shows the corresponding average unit cost for CO₂ capture, transport, and storage (Fig. 6). Three steel-sector capture targets are developed for this analysis: full capture (90%), medium capture (60%), and low capture (30%), which describes the desired CO₂ reduction targets that can be achieved through deploying CCS in China's steel sector.

As the capture target increases, longer pipelines will need to be built to transport captured CO₂ to find suitable CO₂ sinks. In the low-capture scenario, cost-optimized pipeline network will be built preferentially in large CO₂ sources with suitable CO₂ sinks nearby (Fig. 5). With increasing capture targets, the expansion of the pipeline network extends to other areas. However, for China's steel sector, there is an obvious spatial mismatch between CO₂ sources and potential CO₂ sinks, which mean that much of capture CO₂ need to be transported longer distances to find suitable CO₂ sinks. This phenomenon is notably observed in the coastal areas of southeastern and southern China, which hosts large CO₂ sources but has limited available CO₂ sinks nearby. Consequently, in the full-capture scenario, the application of CCS in these areas necessitates longer pipelines to transport captured CO₂ to western sinks.

For higher capture targets, the average unit cost of CCS will increase significantly, of which the CO₂ transportation cost will dominate the cost structure. In the full-capture scenario, the CO₂ transportation cost constitutes 54% of the average unit cost, surpassing the 40% of the capture cost (Fig. 6). This departure from conventional findings, which often emphasize the capture cost as the main expense of CCS in China. This results from the obvious spatial mismatch observed between CO₂ sources and potential CO₂ sinks in China's steel sector. As the cost surface presented in Fig. 3, eastern coastal areas with large CO₂ emissions tend to incur higher pipeline construction costs due to densely populated areas and highly developed land use. Pursuing higher capture targets necessitates the construction of longer pipelines to transport captured CO₂ from eastern areas to western regions to find suitable CO₂ sinks. Consequently, this spatial mismatch results in a significant upswing in CO₂ transportation costs as the capture target increases.

3.3. Regional disparities on the average unit cost of CCS in three selected CO₂ cluster

Under the cost-optimized purpose, large CO₂ sources with suitable CO₂ sinks nearby will be chosen preferentially to build CO₂ pipeline network. To get a deeper look at the regional disparities of these CO₂

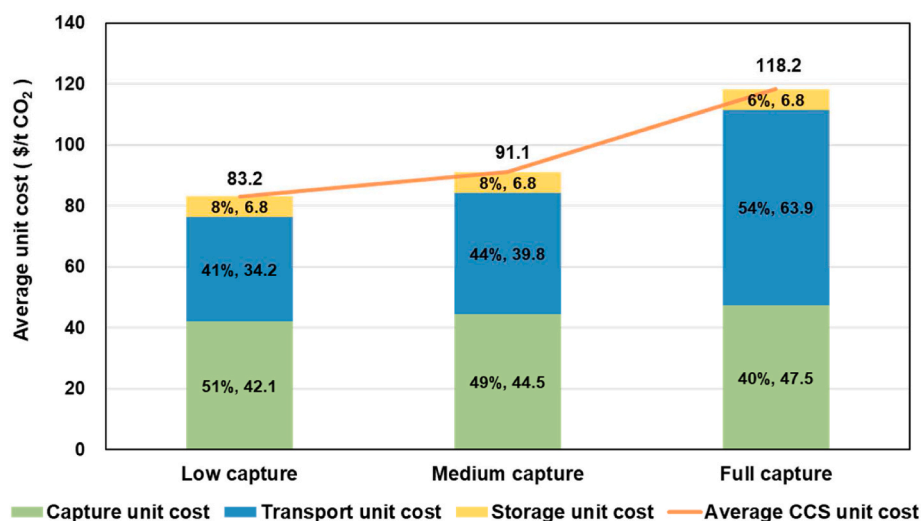


Fig. 6. The trend and composition of the average unit cost of CCS under three capture targets.

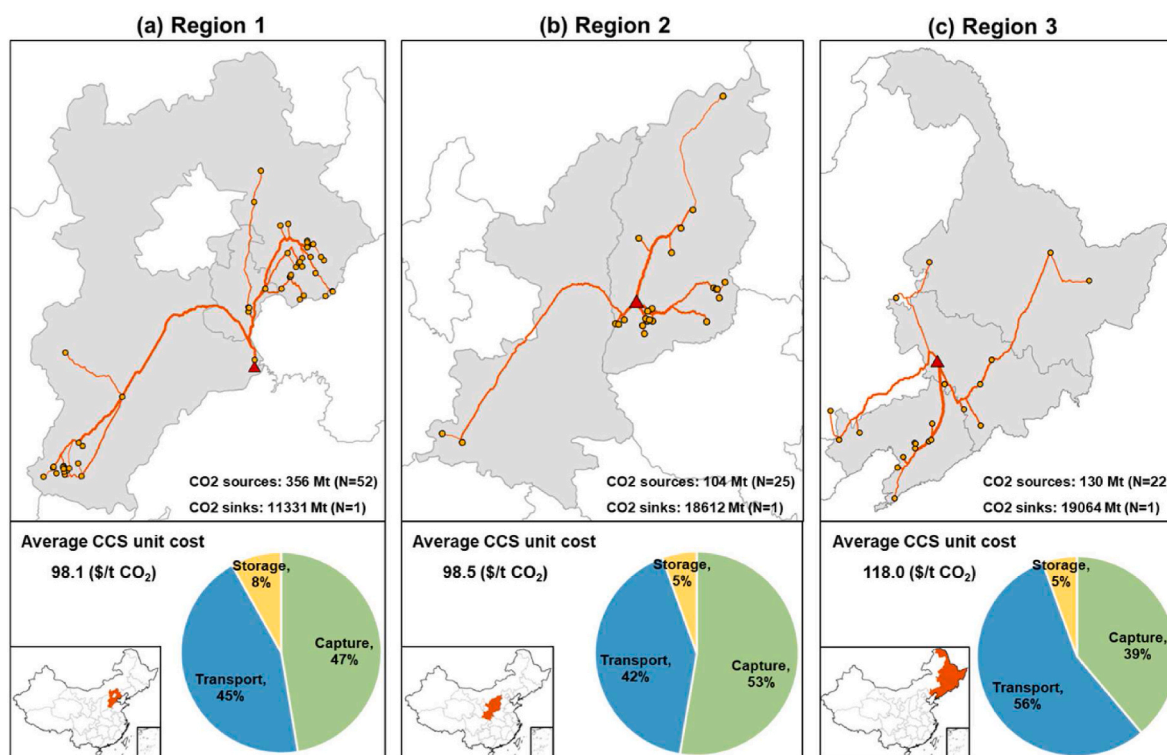


Fig. 7. Pipeline network and average unit cost of CCS in three selected CO₂ clusters under the full-capture scenario.

clusters, this study selected three large CO₂ clusters, presenting the map of pipeline network and the composition of the average unit cost of CCS (Fig. 7). Notably, the average unit cost of CCS for these three CO₂ clusters are all lower than the national average cost. In such regions characterized by large CO₂ sources and suitable CO₂ sinks nearby, the capture of more CO₂ coupled with the construction of shorter pipelines results in a significant reduction in the average unit cost of CCS. In addition, from the perspective of CO₂ sinks, the larger the surrounding CO₂ sources and the closer they are to the sink, the lower the cost. This is clearly caused by the effect of economies of scale, which can be verified by the comparison between region 1 and region 3.

4. Discussion

This study highlights the considerable additional expenses associated with implementing CCS in China's steel sector. In particular, as capture targets increase, CO₂ transportation costs dominates the cost structure of the average unit cost of CCS. The construction of extensive pipelines for transporting CO₂ presents challenges in terms of infrastructure investment. This section further discusses the uncertainties and various issues associated with the application of CCS in China's steel sector. Key considerations include the regions selected for priority CCS deployment, the uncertainty surrounding CO₂ geological storage, the potential viability of offshore CO₂ storage, and the prospect of utilizing captured CO₂ to mitigate costs.

4.1. Large CO₂ sources with suitable CO₂ sinks nearby can be selected for priority CCS deployment

As analyzed in this study, cost-optimized pipeline network is preferentially to be built in regions with large CO₂ sources and suitable CO₂ sinks nearby. Deploying CCS in such regions allows more CO₂ to be captured and transported shorter distances, leading to lower CCS costs. This favorable condition facilitates the prioritized deployment of CCS in these regions. Moreover, the availability of basins will determine the key locations where China's CCS projects will be prioritized for deployment. China's CO₂ sources are mainly distributed in the eastern region, but available basins are poor and mostly in the western region. If a basin suitable for CO₂ geological storage happens to be close to the east and has a large CO₂ emission source nearby, then this basin will become one of the potential options for CCS priority deployment. According to Figs. 4 and 7, such potential CO₂ source-sink pairs include Ordos Basin—Shanxi Province, Songliao Basin—Liaoning Province, and Bohai Bay Basin—Hebei Province.

4.2. The uncertainty of CO₂ geological storage may lead to higher CCS costs

This study strives to mitigate uncertainties associated with the assessment of CO₂ storage capacities and costs of deep saline aquifers. To achieve this, a dynamic model was employed to evaluate the CO₂ storage capacities and costs of deep saline aquifers, incorporating both static geological parameters and dynamic CO₂ injection rates (Midleton et al., 2020b). Subsequently, the Monte Carlo method was employed to assess the probability distribution of CO₂ storage capacity and cost, accounting for variations in input parameters spanning from minimum to maximum feasible values that were found in previous studies (Fig. 8). Ultimately, the mean value derived from this probability distribution was utilized as the representative CO₂ storage capacity and storage cost for the source-sink matching analysis. Notably, the estimated CO₂ storage capacity in this study significantly surpasses the current total CO₂ emissions from China's steel sector, thereby affirming its adequacy to fulfill the CO₂ geological storage requirements in that sector.

However, it is crucial to acknowledge that substantial uncertainties persist in the geological storage of CO₂ due to the limited amount of field experimental data. Many previous studies express optimism regarding China's CO₂ geological storage opportunities, asserting that they can meet all requirements. However, insufficient attention has been directed towards researching the CO₂ geological storage potential within decarbonization models. The deep saline aquifers, situated thousands of meters below the basin, may not store as much CO₂ as anticipated. The uncertainty associated with CO₂ geological storage hinges on numerous variables, encompassing uncertainties in CO₂ injection rates over extended time scales, the geological effectiveness of CO₂ sealing, and the potential impact of seismic activity within earthquake zones on CO₂ storage (Lane et al., 2021). Mitigating these uncertainties necessitates an increase in on-site tests, such as extending the testing of injection wells and incorporating more geophysical field modeling (Ren et al., 2023; Gan et al., 2021). Upon identifying a suitable CO₂ storage site, the development of CCS infrastructure will naturally evolve in tandem with the selected site. Additionally, new estimates of CO₂ storage potential can be introduced for macro-level decarbonization modeling. For example, Neil et al. (Grant et al., 2022) proposed that the maximum historical rate of oil & gas production can serve as the indicator of the realistic CO₂ injection rate in a region.

4.3. Offshore CO₂ storage may be a promising way for China's CCS projects

As estimated in this study, for China's coastal areas characterized by substantial CO₂ sources but limited nearby onshore CO₂ storage sites,

the implementation of CCS necessitates long CO₂ pipelines and higher transportation costs to transport captured CO₂ to nearby CO₂ sinks. However, in close proximity to China's coastal regions, offshore basins boast extensive seafloor geological saline aquifers, offering a potential solution for CO₂ storage (Zhou et al., 2018). In this way, captured CO₂ can be transported through shorter pipelines or directly by maritime vessels to offshore storage sites, facilitating subsequent injection and storage. Presently, offshore CO₂ transportation and storage in China are undergoing initial phases of laboratory research and small-scale experimentation, lagging behind the more advanced progress of onshore CCS development (Zhou et al., 2018). For example, CO₂ transportation pipelines necessitate resilience to high-pressure submarine conditions and corrosion from seawater; the intricate submarine environments impose elevated demands on CO₂ injection technologies and equipment. In the future, if offshore CO₂ storage is of high technology readiness and cost-competitive with onshore CO₂ storage, it may emerge as a promising avenue for China's coastal areas abundant in CO₂ sources but possessing limited suitable onshore CO₂ storage options.

4.4. Promoting the utilization of captured CO₂ can be an effective way to reduce costs and get profits for China's steel sector

The costs associated with CCS are anticipated to pose a significant financial burden on steel companies, given the inherent cost sensitivity of steel as a commodity. As estimated in this study, the average unit costs of CCS under low (30%), medium (60%), full capture (90%) targets are 83, 91, and 118 \$/t CO₂, respectively. Then, the production cost of crude steel stands at approximately 500\$/t crude steel. Assuming a direct CO₂ emission of 1.4 tons per ton of crude steel produced (Lin et al., 2023), the integration of CCS will increase the production cost of crude steel by 23%–33% to between 616 and 665 \$/t crude steel. Consequently, advocating for the comprehensive utilization of captured CO₂ emerges as a pivotal strategy to appreciably mitigate the financial burden on steel companies. For instance, the promotion of captured CO₂ for enhanced oil recovery (CO₂-EOR) stands out as a promising avenue, CO₂ mineralization for the synthesis of building materials or conversion to other long-lived products may similarly yield tangible benefits for steel companies, concurrently obviating transportation costs and mitigating uncertainties associated with storage costs.

4.5. Both CCS and hydrogen face huge uncertainties in costs and technical routes

In addition to CCS, hydrogen-based steelmaking is regarded as a pivotal decarbonization option for China's steel sector. To compare it with the CCS, this study conducts a simple cost estimation of the main technical routes for hydrogen steelmaking, as shown in Fig. 9.

It is evident that neither option holds a significant cost advantage. The cost of hydrogen-based steelmaking is estimated at 600–650 \$/t steel, while CCS stands at 616–655 \$/t steel as estimated in section 4.4. Furthermore, both technologies face huge uncertainties in technical route selection. CCS necessitates the explore of high-efficiency, cost-effective CO₂ capture techniques and the identification of suitable CO₂ geological storage sites or the comprehensive utilization of captured CO₂. Hydrogen-based steelmaking technology demands a fundamental overhaul of the entire steel production process and requires substantial upfront investment, alongside a certain risk of asset stranding. In general, both CCS and hydrogen-based steelmaking technology are at the early stage of technical readiness. Progressing from laboratory experiments to large-scale commercial deployment necessitates extensive further research and engineering demonstration projects, to enhance technological readiness for the eventual deep decarbonization of China's steel sector.

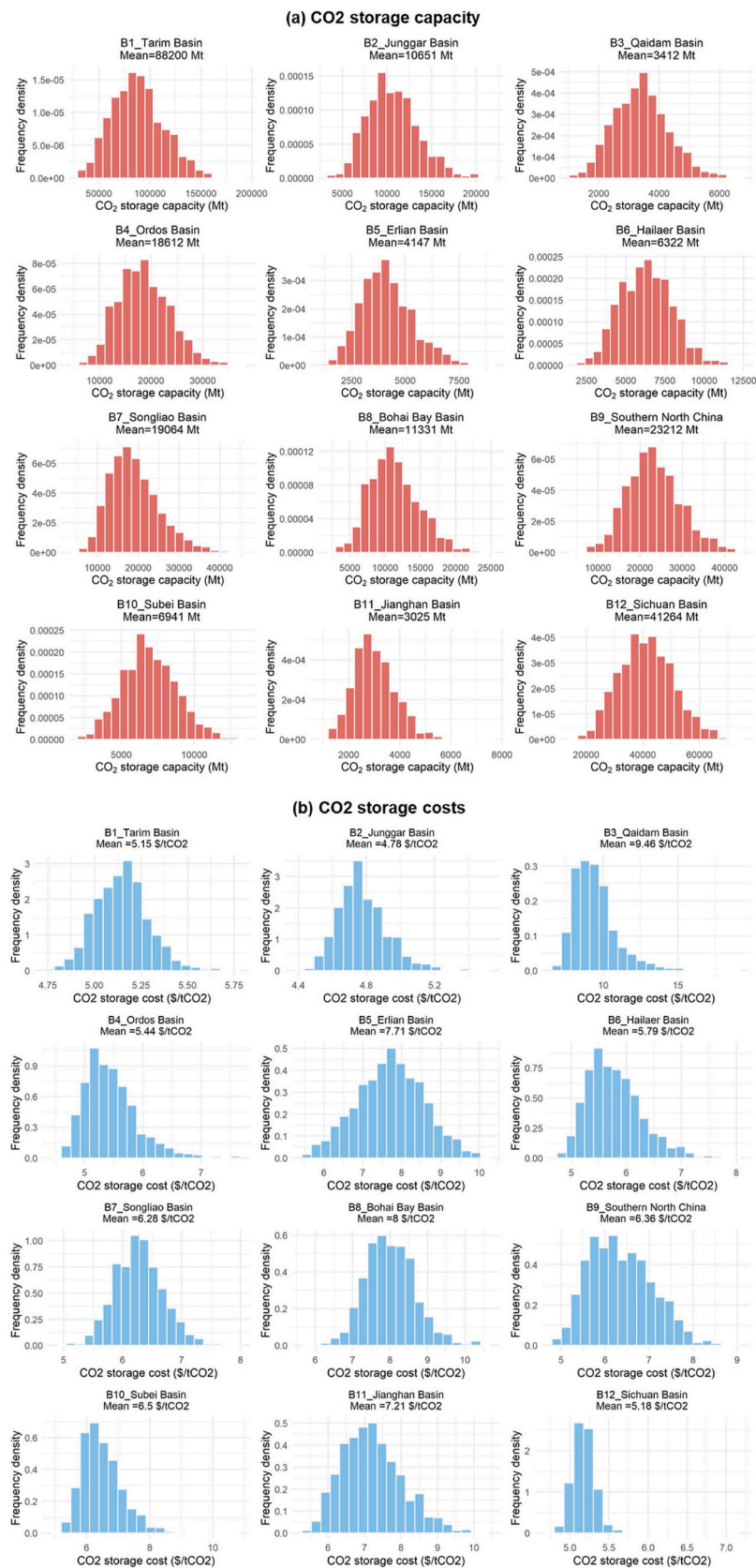


Fig. 8. The uncertainty analysis of CO₂ storage capacities and costs of 12 deep saline aquifers in China (The results of uncertainty analysis are presented using frequency density histogram. In each sample ($N = 1000$), the area of the bar corresponding to the interval (m, n) denotes the frequency, with the collective sum of all bar areas equating to 1. The mean value is calculated by multiplying the midpoint of the interval (m, n) by their respective frequencies and then summing them up.).

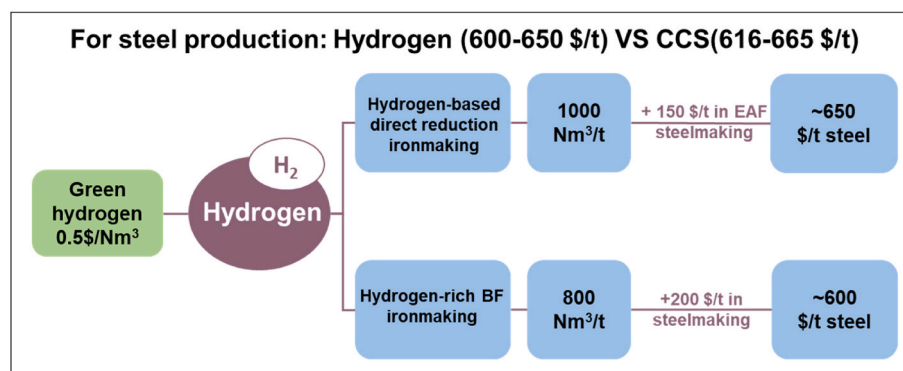


Fig. 9. Cost comparison between CCS and hydrogen in China's steel sector based on a simple cost estimation of hydrogen-based steelmaking.

5. Conclusions

This study delved into the CO₂ reduction potential and associated costs through a comprehensive source-sink matching analysis, with the objective of offering early engineering guidance for the deployment of CCS in China's steel sector. Firstly, a comprehensive dataset of CO₂ sources was developed, encompassing 227 BF/BOF steel plants, featuring capturable CO₂ amounts, CO₂ capture costs, and location coordinates. Secondly, an assessment of 12 deep saline aquifers was conducted to develop preliminary estimates of CO₂ storage capacities and costs. Finally, cost-optimized pipeline networks were delineated under three distinct capture targets, using the open-source optimization tool, SimCCS.

This study reveals that:

- There is an obvious spatial mismatch between current steel-sector CO₂ sources and major CO₂ storage sites of China. CO₂ sources from steel plants in China are primarily situated in the eastern region, while the main CO₂ storage sites are located in the western part of the country.
- The average unit costs of CCS under low (30%), medium (60%), full capture (90%) targets are 83, 91, and 118 \$/t CO₂, respectively. Given current production costs for crude steel in China, implementing CCS represents 20% additional costs for steel companies.
- As the capture target increases, CO₂ transportation costs will denominate the total CCS costs. In the full-capture scenario, the CO₂ transportation cost constitutes 54% of the total CCS costs, surpassing 40% of the capture cost. This is attributed to the spatial mismatch between CO₂ sources and sinks, leading to the construction of longer pipelines when pursuing higher capture targets.

Based on the results of this study, the main policy implication for the deployment of CCS in China's steel sector is to increase R&D investment and promote engineering demonstrations for technical advancement and cost mitigation, aiming to foster technological readiness for the eventual deep decarbonization of China's steel sector. Firstly, it is recommended to prioritize CCS demonstration projects in regions with concentrated CO₂ clusters and suitable CO₂ storage basins nearby, such as Shanxi, Liaoning, and Hebei. This can offer valuable practical experience for subsequent large-scale applications. Additionally, steel producers are encouraged to explore the multiple utilization of captured CO₂ for generating profits, such as converting CO₂ to long-lives products. Furthermore, the study emphasizes the offshore CO₂ storage as a promising solution for China's coastal areas, potentially mitigating CO₂ transportation cost limitations associated with onshore CO₂ storage.

In the future, more source-sink matching studies can be carried out at the regional level to enhance the results of this study, especially in the selected areas for prioritizing deployment of CCS. This necessitates the availability of more refined datasets of CO₂ sources and sinks, as well as

higher-resolution GIS maps of selected regions. Moreover, additional research avenues can be explored based on the foundation laid by this study:

- Technical aspect. Despite the progress in CO₂ capture technologies, most have only undergone small-scale engineering experiments in certain power plants. Significant uncertainties still exist regarding their applicability in steel plants and the optimal selection of technological routes. Hence, further laboratory studies and plant-level demonstration projects are necessary for technological advancement and cost reduction. Furthermore, conducting additional on-site experiments to assess the suitability of CO₂ geological storage sites is essential. Moreover, more on-site experiments of CO₂ storage sites are essential for identifying suitable CO₂ geological storage sites.
- Socio-economic aspects. The siting of CO₂ pipeline construction can be influenced significantly by socio-economic factors, such as imbalances in regional development and public acceptance of large-scale CO₂ transportation. Incorporating these socio-economic factors into the construction of cost surfaces is vital for informed decision-making of CO₂ pipeline construction.
- Financial aspect. Presently, the application of CCS poses a considerable financial burden to steel companies. Hence, exploring the effects of financial instruments, such as targeted subsidies, carbon taxes, or carbon market mechanisms becomes crucial to incentivize enterprises to reduce CO₂ emission and expedite the deployment of CCS.

CRediT authorship contribution statement

Yuancheng Lin: Writing – original draft, Visualization, Validation, Investigation, Formal analysis. **Tubagus Aryandi Gunawan:** Writing – review & editing, Supervision, Software. **Cecelia Isaac:** Software. **Hongxi Luo:** Writing – review & editing, Supervision. **Fangwei Cheng:** Supervision. **Eric D. Larson:** Writing – review & editing, Resources, Conceptualization. **Chris Greig:** Supervision. **Linwei Ma:** Supervision, Project administration, Funding acquisition. **Zheng Li:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared our data at the attached file.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2024.142280>.

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